Theory of Computation

Regular Languages and Regular Expressions

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Single Final State for NFAs and DFAs

- Any Finite Automaton (NFA or DFA) can be converted to an equivalent NFA with a single final state.

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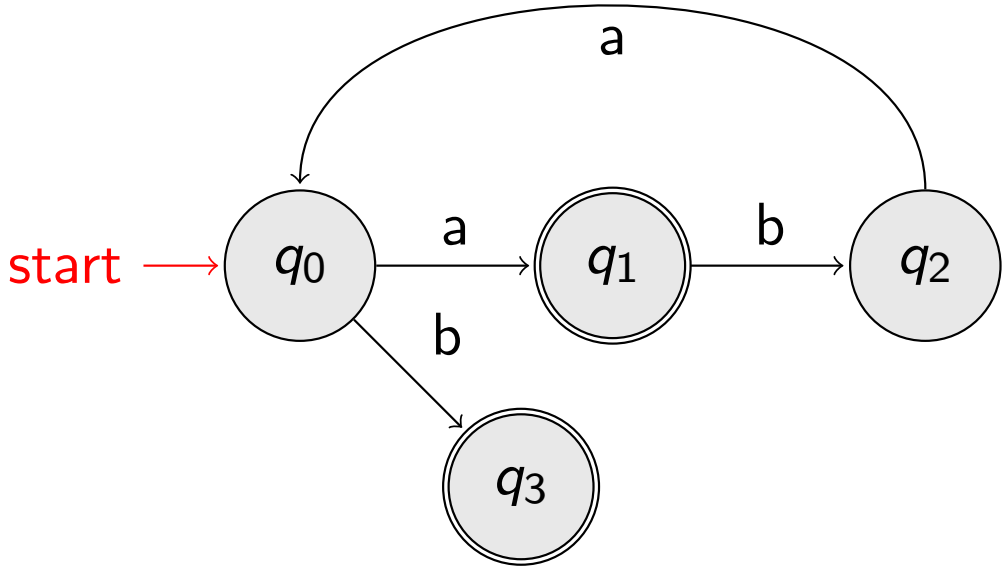
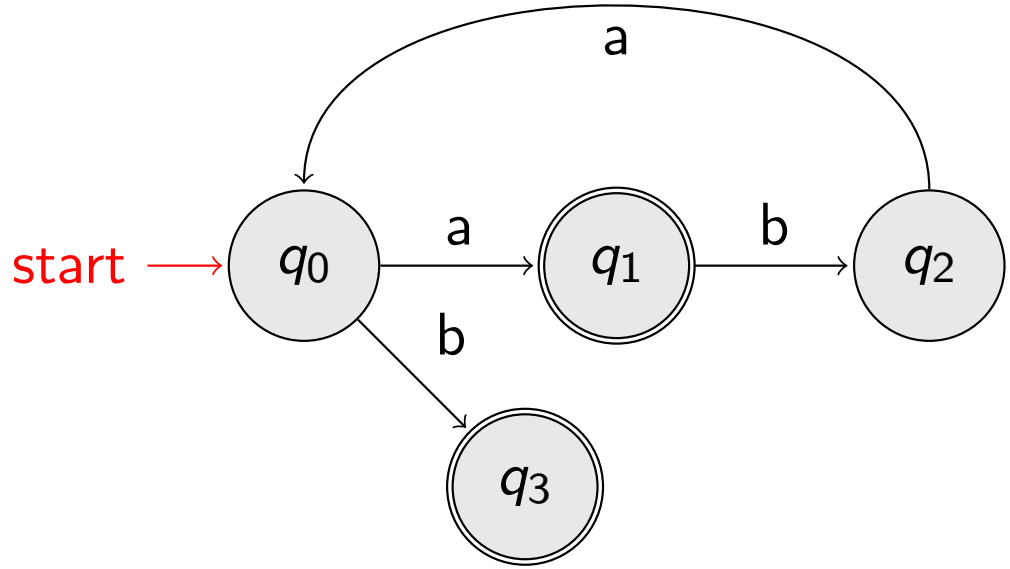
Single Final State for NFAs and DFAs (Example:)



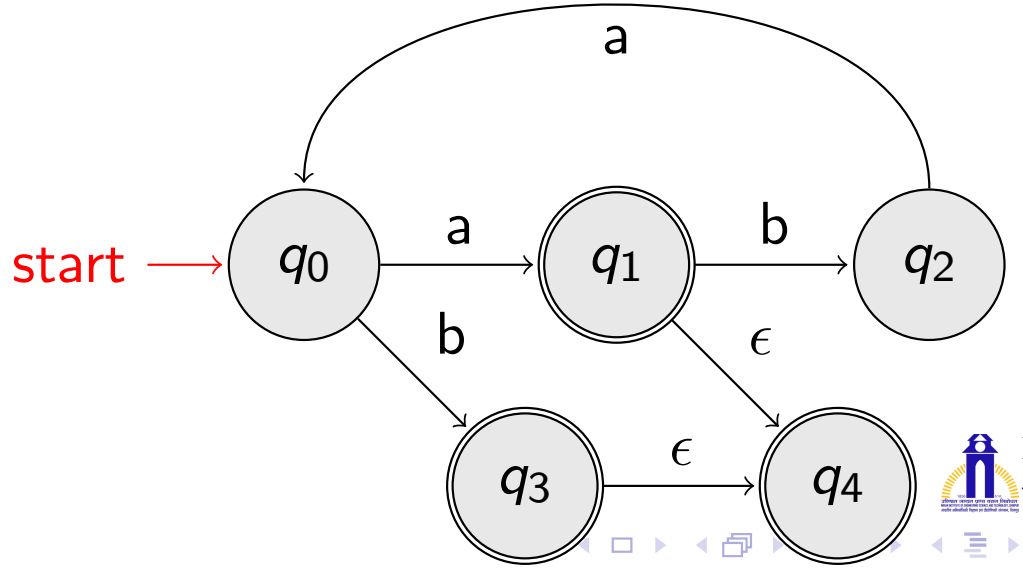
- **NFA**

Single Final State for NFAs and DFAs (Example:)

NFA



Equivalent NFA



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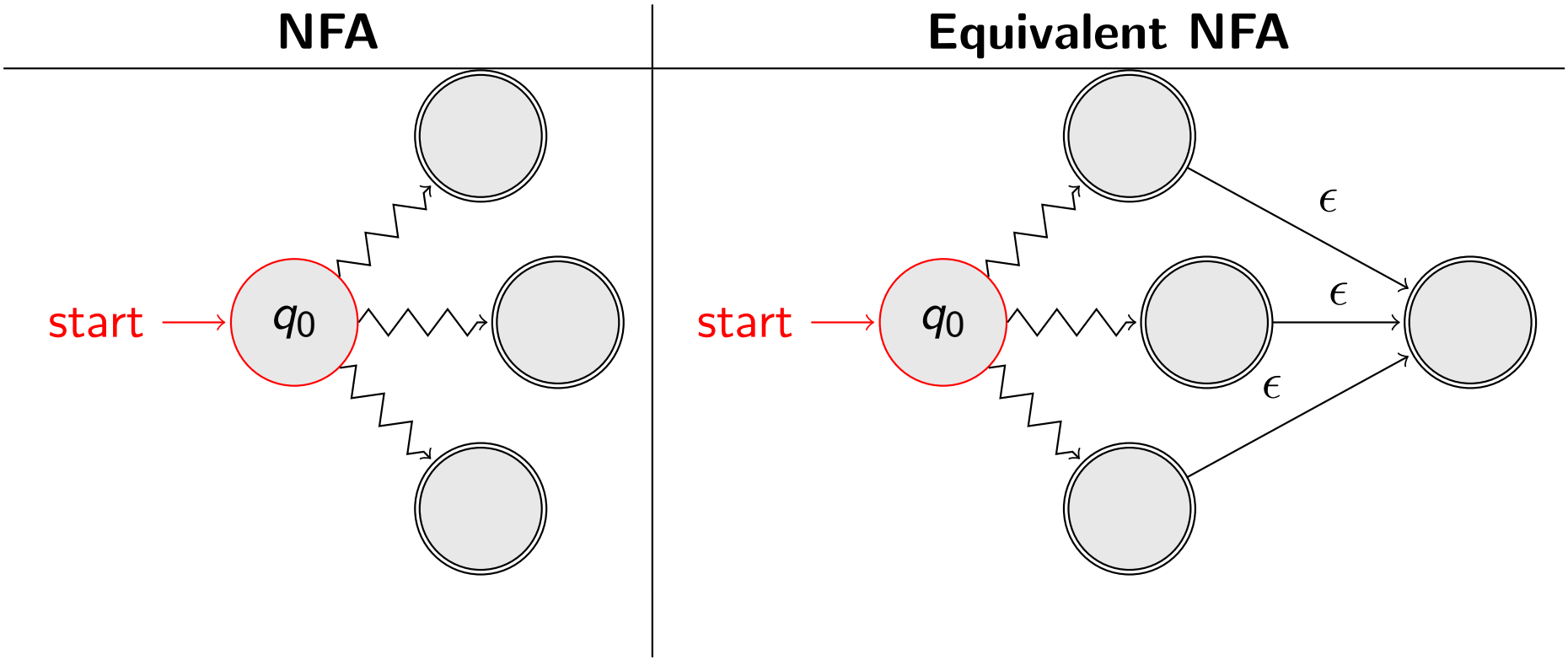
Single Final State for NFAs and DFAs (In general)



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Some Properties of Regular Languages

- For regular languages L_1 and L_2 we will prove that:

$$\left.\begin{array}{l} \textbf{Union: } L_1 \bigcup L_2 \\ \textbf{Concatenation: } L_1 L_2 \\ \textbf{Star: } L_1^* \end{array}\right\} = \{\textbf{Are Regular Languages}\}$$

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- We say:
  Regular languages are **closed** under

$$\left.\begin{array}{l} \textbf{Union: } L_1 \bigcup L_2 \\ \textbf{Concatenation: } L_1 L_2 \\ \textbf{Star: } L_1^* \end{array}\right\}$$

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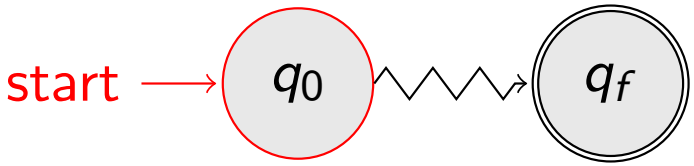
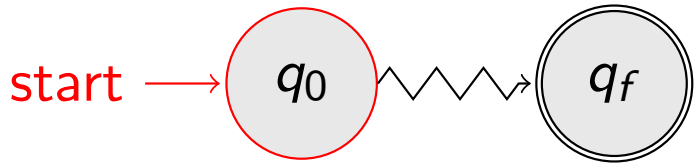
Some Properties of Regular Languages

- Properties:

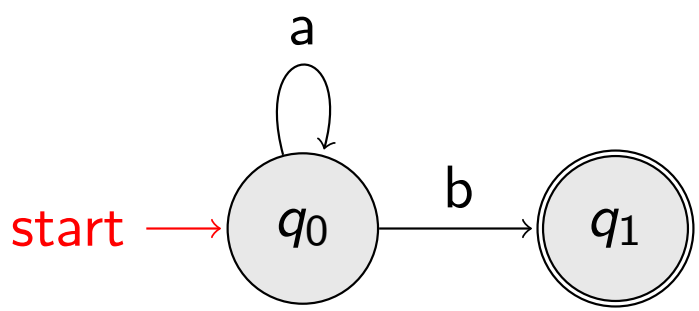
Regular language, L_1	**Regular language,** L_2
$L(M_1) = L_1$	$L(M_2) = L_2$
NFA M_1	NFA M_2
	
Single Final State	Single Final State

Some Properties of Regular Languages

- **Example:**

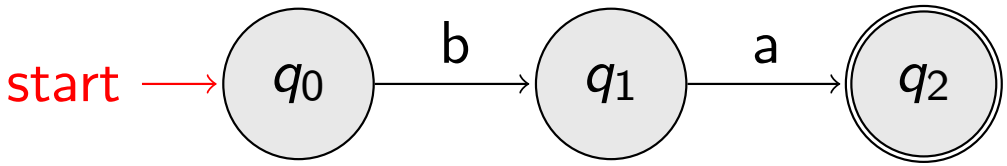
$L_1 = \{a^n b\}$

M_1



$L_2 = \{ba\}$

M_2



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Some Properties of Regular Languages

- **NFA for** $L_1 \bigcup L_2$

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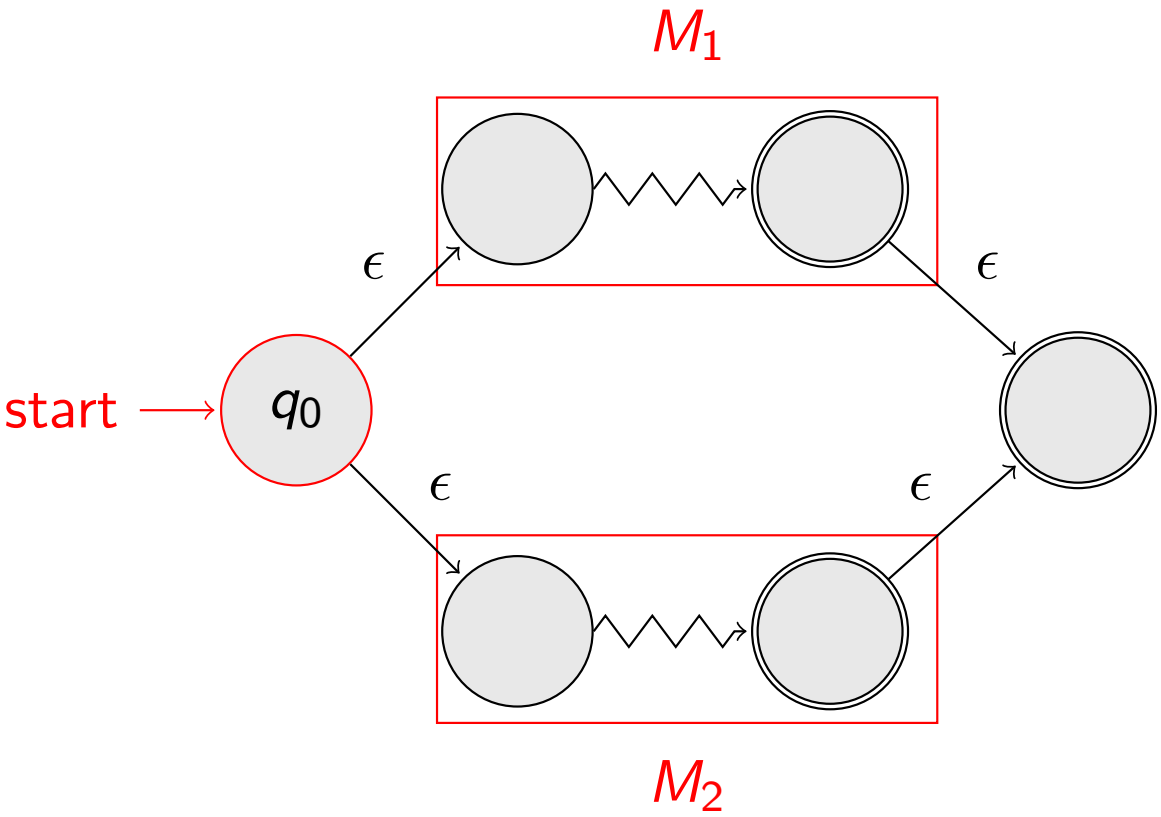

Some Properties of Regular Languages

- **Example:** NFA for $L_1 \bigcup L_2 = \{a^n b\} \bigcup \{ba\}$

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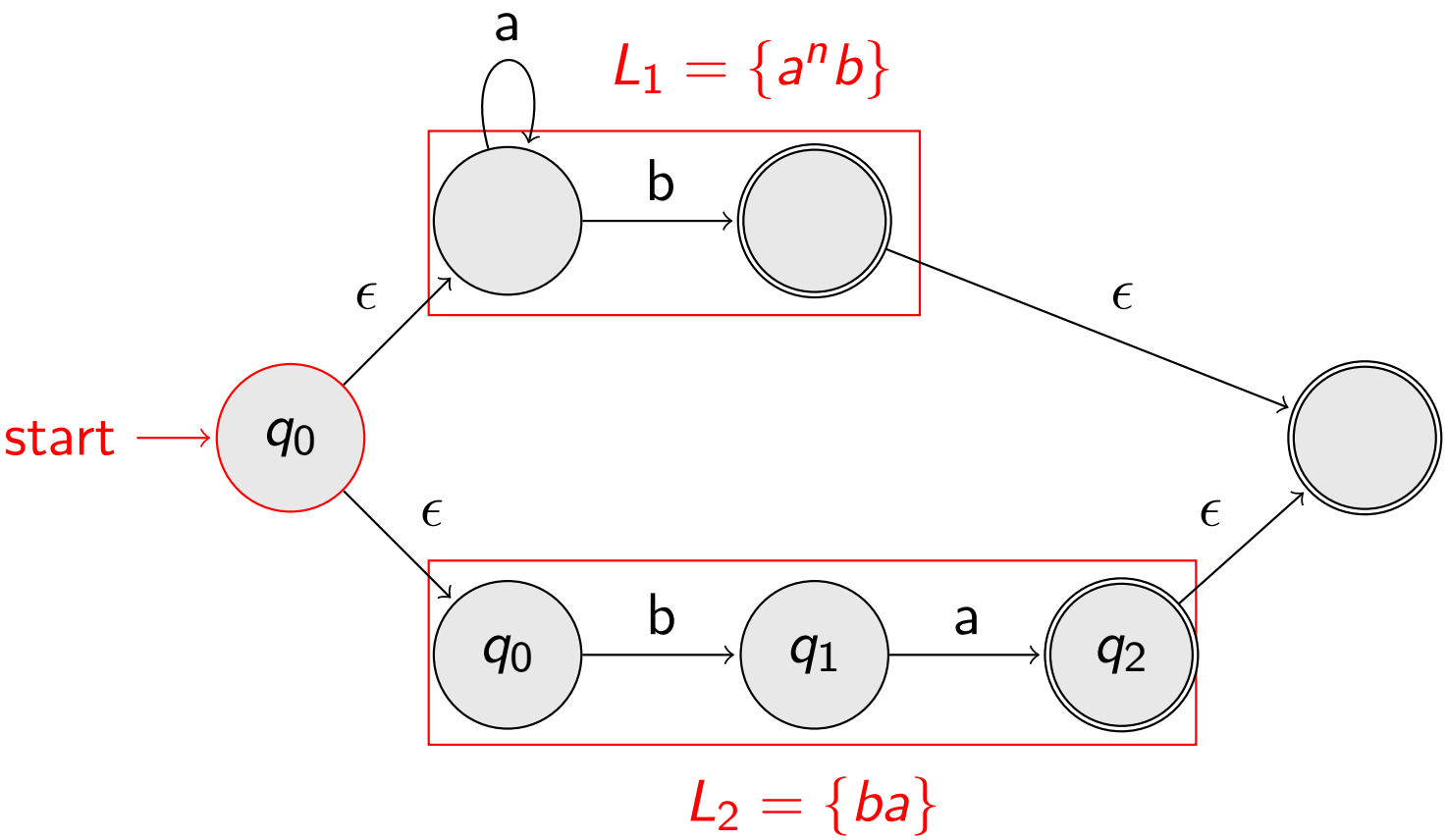

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Some Properties of Regular Languages

- **Concatenation:**
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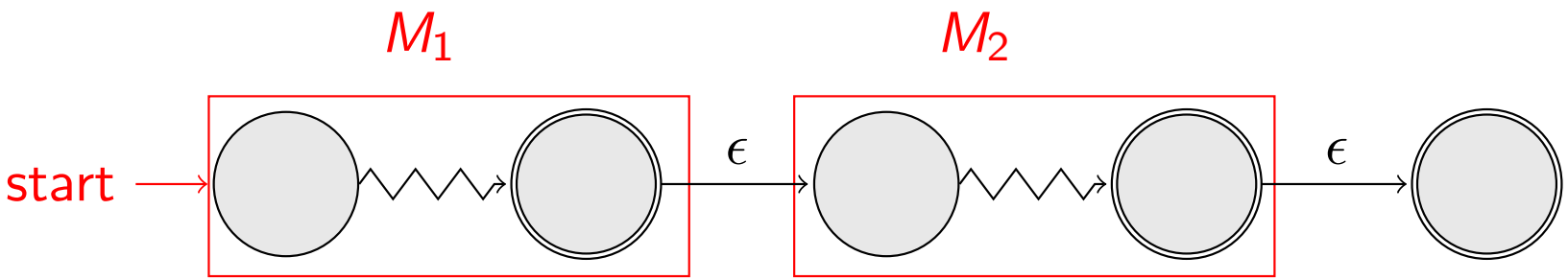

Some Properties of Regular Languages

- **Example:** NFA for $L_1L_2 = \{a^n b\}\{ba\} = \{a^n bba\}$

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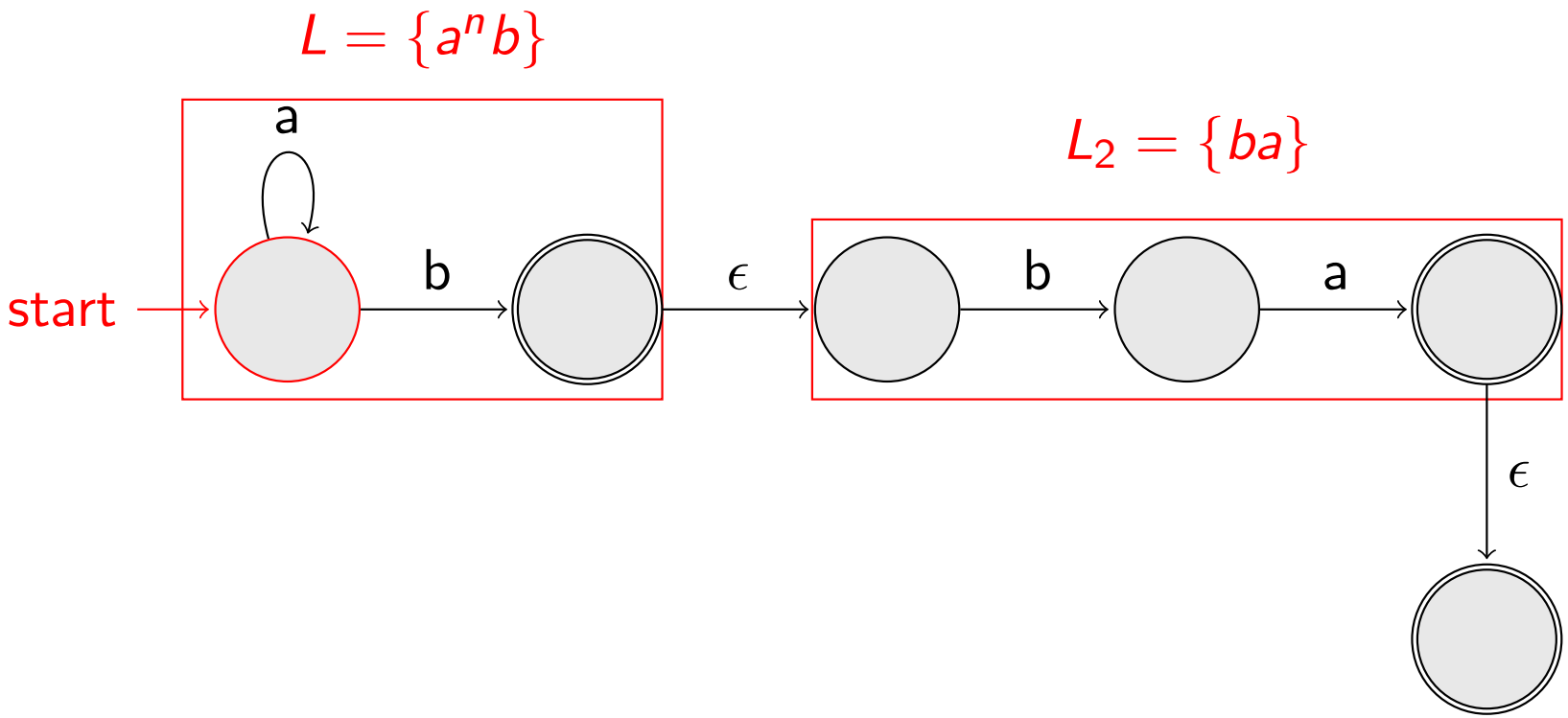

Some Properties of Regular Languages

- **Star Operation:** NFA for L_1^*

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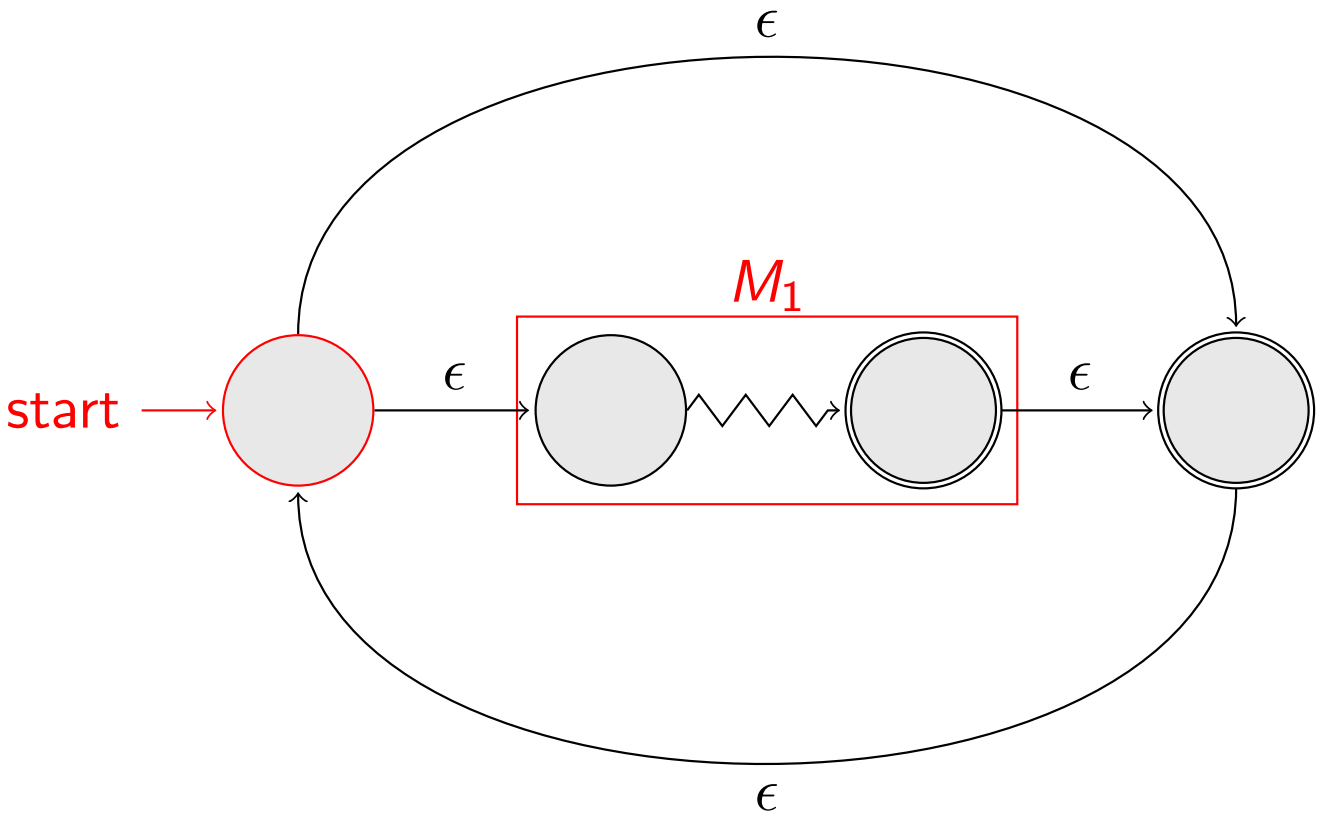

Some Properties of Regular Languages

- **Example:** NFA for $L_1^* = \{a^n b\}^*$

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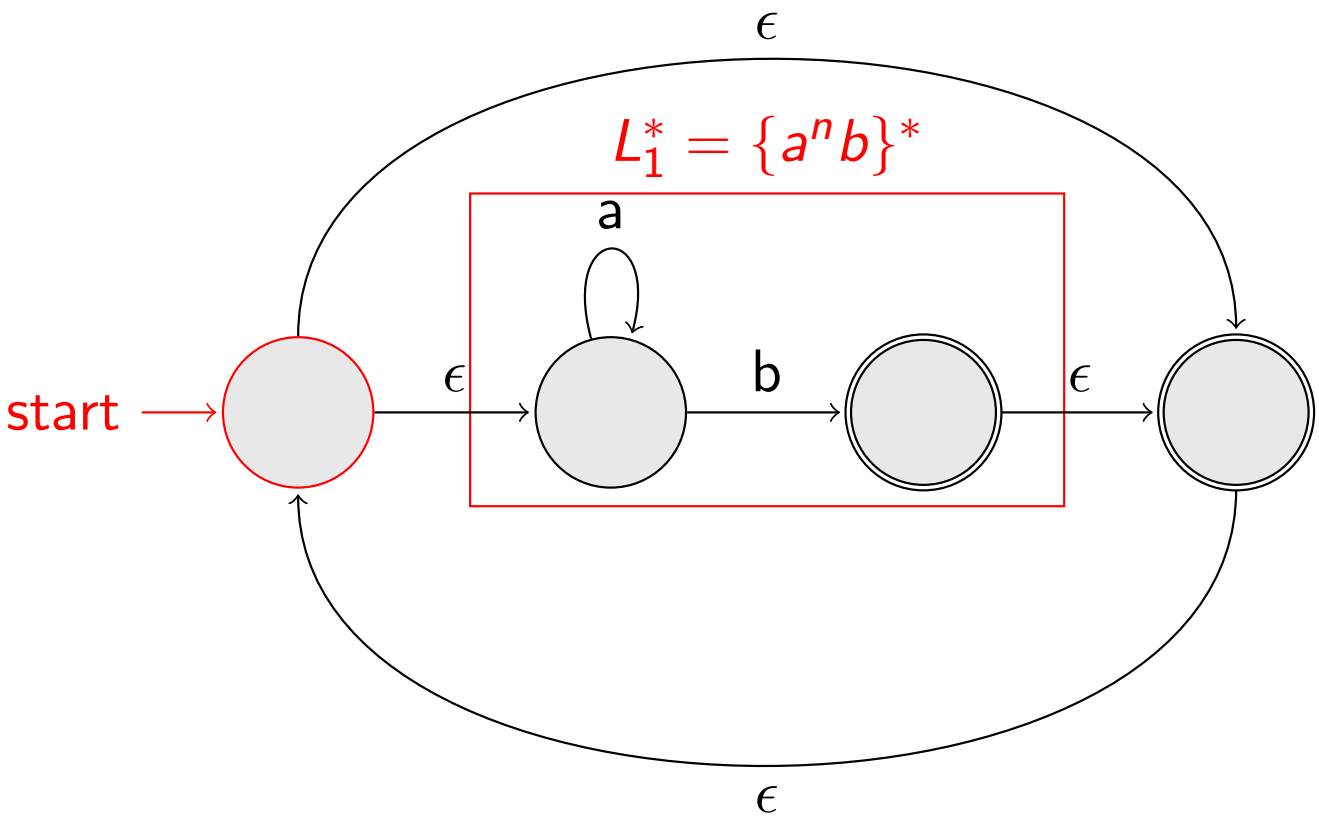

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Regular Expressions

- The Language accepted by **Finite Automata** are easily described by simple expressions called **Regular Expressions**.

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Regular Expressions

- The Language accepted by **Finite Automata** are easily described by simple expressions called **Regular Expressions**.
- **Regular Expression** is a way to represent a language.
- **Regular expressions** describe **Regular language**.
- **Example:** $(a + bc)^*$
  describes the language
  $\{a, bc\}^* = \{\lambda, a, bc, aa, abc, bca, \ldots\}$

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Regular Expression

- **Recursive Definition:**
  - Primitive regular expressions: Φ, λ, α

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$$\left.\begin{array}{l} r_1 + r_2 \\ r_1.r_2 \\ r_1^* \\ (r_1) \end{array}\right\} \text{Are regular expressions}$$

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  - A regular expression: $(a + bc)^*(c + \Phi)$

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  - A regular expression: $(a + bc)^*(c + \Phi)$
  - Not a regular expression: (a+b+)

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Languages of Regular Expressions

- L(R): language of regular expression, r
  Example: $L((a+b+c)^*) = \{\lambda, a, bc, aa, abc, bca, \dots\}$

Languages of Regular Expressions

- L(R): language of regular expression, r
  Example: $L((a+b+c)^*) = \{\lambda, a, bc, aa, abc, bca, \dots\}$
- For primitive regular expressions:
  - $L(\Phi) = \Phi$
  - $L(\lambda) = \{\lambda\}$
  - $L(a) = \{a\}$
  - $L(r_1 + r_2) = L(r_1) \bigcup L(r_2)$
  - $L(r_1.r_2) = L(r_1)L(r_2)$
  - $L(r_1^*) = (L(r_1))^*$
  - $(L(r_1)) = L(r_1)$

Languages of Regular Expressions

- **Example:**
  Regular Expression: $(a+b).a^*$
  $L((a+b).a^*) = L(a+b)L(a^*) = (L(a) \bigcup L(b))L(a^*) =$
  $(\{a\} \bigcup \{b\}\{(a)\}^* = \{a, b\}\{\lambda, a, aa, aaa, \ldots\} =$
  $\{a, aa, aaa, \ldots, b, bb, bbb, \ldots\}$

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- **Regular Expression:** $r = (a+b)^*(a+bb)$
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  $L(r) = \{a^{2n}b^{2m}b : n, m \geq 0\}$

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- **Regular Expression:** $r = (0+1)^*00(0+1)^*$
  $L(r) = \{$all strings with at least two consecutive $0\}$

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- **Regular Expression:** $r = (0+1)^*00(0+1)^*$
  $L(r) = \{$all strings with at least two consecutive 0$\}$
- **Regular Expression:** $r = (1+01)^*(0+\lambda)$
  $L(r) = \{$all strings without two consecutive 0$\}$

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Equivalent Regular Expressions

- Regular expressions r_1 and r_2 are equivalent if $L(r_1) = L(r_2)$
- **Example:** L = { all strings without two consecutive 0 }
  $r_1 = (0 + 01)^*(0 + \lambda)$
  $r_2 = (1^*011^*)^*(0 + \lambda) + 1^*(0 + \lambda)$
  $L(r_1) = L(r_2) = L \implies r_1$ and r_2 are equivalent regular expression.

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